

HDOC - Day 27

Menu-Driven Pyramid and Diamond Patterns

Algorithm

1. Start and display the menu for five pyramid/diamond patterns and Exit.
2. Read the menu choice and use switch-case to call the selected parameterless function.
3. Read n, the required number of rows.
4. Use nested loops to print leading spaces and the required stars/numbers for each row.
5. For a diamond, print the upper pyramid followed by the lower inverted part.
6. Repeat the menu until Exit is selected, then stop.

Test Case Table

Test	Choice	n	Expected output
1	1	4	Full pyramid
2	2	4	Inverted pyramid
3	3	4	Diamond (7 rows)
4	4	4	Palindromic number pyramid
5	5	4	Repeated-number pyramid
6	6	-	Exit message

C Program

```
#include <stdio.h>
void fullPyramid(); void invertedPyramid(); void diamond();
void palindrome(); void repeated();
int main(){int ch; do{
    printf("\n1.Full pyramid 2.Inverted pyramid 3.Diamond\n");
    printf("4.Palindromic number pyramid 5.Repeated-number pyramid 6.Exit\n");
    printf("Enter choice: "); scanf("%d",&ch);
    switch(ch){case 1:fullPyramid();break;case 2:invertedPyramid();break;
    case 3:diamond();break;case 4:palindrome();break;case 5:repeated();break;
    case 6:printf("Exiting...\n");break;default:printf("Invalid choice\n");}
}while(ch!=6);return 0;}
void fullPyramid(){int n,i,j;printf("Enter rows: ");scanf("%d",&n);
    for(i=1;i<=n;i++){for(j=1;j<=n-i;j++)printf(" ");for(j=1;j<=2*i-1;j++)printf("* ");printf("\n");}}
void invertedPyramid(){int n,i,j;printf("Enter rows: ");scanf("%d",&n);
    for(i=n;i>=1;i--){for(j=1;j<=n-i;j++)printf(" ");for(j=1;j<=2*i-1;j++)printf("* ");printf("\n");}}
void diamond(){int n,i,j;printf("Enter rows: ");scanf("%d",&n);
    for(i=1;i<=n;i++){for(j=1;j<=n-i;j++)printf(" ");for(j=1;j<=2*i-1;j++)printf("* ");printf("\n");}
    for(i=n-1;i>=1;i--){for(j=1;j<=n-i;j++)printf(" ");for(j=1;j<=2*i-1;j++)printf("* ");printf("\n");}}
void palindrome(){int n,i,j;printf("Enter rows: ");scanf("%d",&n);
    for(i=1;i<=n;i++){for(j=1;j<=n-i;j++)printf(" ");for(j=1;j<=i;j++)printf("%d ",j);for(j=i-1;j>=1;j--)printf("%d ",j);printf("\n");}}
void repeated(){int n,i,j;printf("Enter rows: ");scanf("%d",&n);
    for(i=1;i<=n;i++){for(j=1;j<=n-i;j++)printf(" ");for(j=1;j<=2*i-1;j++)printf("%d ",i);printf("\n");}}
```

Compilation, Execution & Testing

Test Case 1 - Full pyramid

```
puppie@Anvikas-MacBook-Air Desktop % clang day27.c -o day27
puppie@Anvikas-MacBook-Air Desktop % ./day27
Enter choice: 1
Enter rows: 4
  *
 * * *
* * * * *
* * * * * * *
```

Test Case 2 - Inverted pyramid

```
puppie@Anvikas-MacBook-Air Desktop % ./day27
Enter choice: 2
Enter rows: 4
* * * * *
 * * * *
  * * *
   * *
```

Test Case 3 - Diamond

```
puppie@Anvikas-MacBook-Air Desktop % ./day27
Enter choice: 3
Enter rows: 4
  *
 * * *
* * * * *
* * * * * * *
 * * * *
  * * *
   *
```

Test Case 4 - Palindromic pyramid

```
puppie@Anvikas-MacBook-Air Desktop % ./day27
Enter choice: 4
Enter rows: 4
  1
 1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
```

Test Case 5 - Repeated-number pyramid

```
puppie@Anvikas-MacBook-Air Desktop % ./day27
Enter choice: 5
Enter rows: 4
  1
 2 2 2
3 3 3 3 3
4 4 4 4 4 4 4
```

Test Case 6 - Exit

```
puppie@Anvikas-MacBook-Air Desktop % ./day27
Enter choice: 6
Exiting...
puppie@Anvikas-MacBook-Air Desktop %
```

Result: The menu-driven C program for pyramid and diamond patterns was implemented using switch-case and nested loops and tested successfully.